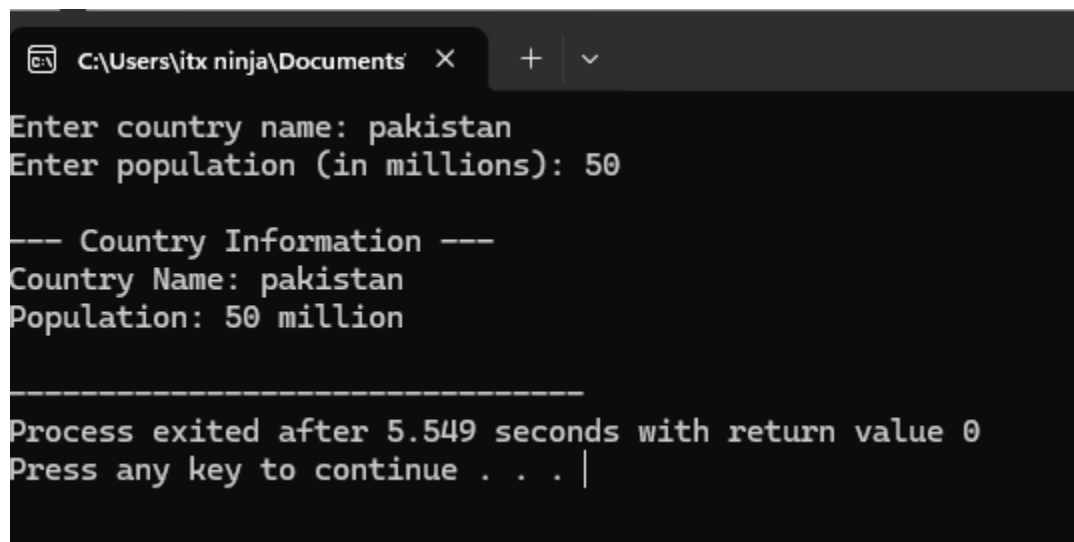


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ROLL NO:SU91-BSRAM-F25-040

LAB3

```
/*task1
#include <iostream>
using namespace std;
struct Country {
    string name;
    float population;
};
int main() {
    Country c;
    cout << "Enter country name: ";
    cin >> c.name;
    cout << "Enter population (in millions): ";
    cin >> c.population;
    cout << "\n--- Country Information ---" << endl;
    cout << "Country Name: " << c.name << endl;
    cout << "Population: " << c.population << " million" << endl;
    return 0;
}
```

TASK1: 



```
Enter country name: pakistan
Enter population (in millions): 50

--- Country Information ---
Country Name: pakistan
Population: 50 million

-----
Process exited after 5.549 seconds with return value 0
Press any key to continue . . . |
```

TASK2:

```

/*task2
#include <iostream>
using namespace std;
struct Student {
    int rollNo;
    float marks[5];
};
int main() {
    Student s;
    cout << "Enter Roll Number: ";
    cin >> s.rollNo;
    cout << "Enter marks of 5 subjects:\n";
    for(int i = 0; i < 5; i++) {
        cout << "Subject " << i + 1 << ": ";
        cin >> s.marks[i];
    }
    cout << "\n--- Student Record ---" << endl;
    cout << "Roll Number: " << s.rollNo << endl;
    cout << "Marks in 5 subjects:\n";
    for(int i = 0; i < 5; i++) {
        cout << "Subject " << i + 1 << ": " << s.marks[i] << endl;
    }
    return 0;
}*/

```

```

C:\Users\itx ninja\Documents
Enter Roll Number: 40
Enter marks of 5 subjects:
Subject 1: 50
Subject 2: 65
Subject 3: 93
Subject 4: 89
Subject 5: 76

--- Student Record ---
Roll Number: 40
Marks in 5 subjects:
Subject 1: 50
Subject 2: 65
Subject 3: 93
Subject 4: 89
Subject 5: 76

-----
Process exited after 12.03 seconds with return value 0
Press any key to continue . . .

```

TASK3:

```

/*task3
#include <iostream>
using namespace std;
struct Car {
    string model;
    string color;
    float price;
};
int main() {
    Car c[5];
    cout << "Enter details of 5 cars:\n";
    for(int i = 0; i < 5; i++) {
        cout << "\nCar " << i + 1 << endl;
        cout << "Enter model: ";
        cin >> c[i].model;
        cout << "Enter color: ";
        cin >> c[i].color;
        cout << "Enter price: ";
        cin >> c[i].price;
    }
    int choice;
    cout << "\nEnter 1 to display most costly car\n";
    cout << "Enter 2 to display all cars\n";
    cout << "Your choice: ";
    cin >> choice;
    if(choice == 1) {
        int maxIndex = 0;
        for(int i = 1; i < 5; i++) {
            if(c[i].price > c[maxIndex].price) {
                maxIndex = i;
            }
        }
        cout << "\n--- Most Costly Car ---\n";
        cout << "Model: " << c[maxIndex].model << endl;
        cout << "Color: " << c[maxIndex].color << endl;
        cout << "Price: " << c[maxIndex].price << endl;
    }
}

```

```
C:\Users\itx ninja\Documents' X + v
Enter model: 2025
Enter color: black
Enter price: 2,00000

Car 3
Enter model: Enter color: grey
Enter price: 50000000

Car 4
Enter model: 2019
Enter color: pink
Enter price: 7800000

Car 5
Enter model: 2015
Enter color: red
Enter price: 500000

Enter 1 to display most costly car
Enter 2 to display all cars
Your choice: 1

--- Most Costly Car ---
Model: ,00000
Color: grey
Price: 5e+007

-----
Process exited after 115.9 seconds with return value 0
Press any key to continue . . . |
```

TASK4:

```

/*task4
#include <iostream>
using namespace std;
struct phone {
    int areaCode;
    int exchange;
    int number;
};
int main() {
    phone myNumber = {212, 767, 8900};
    phone userNumber;
    cout << "Enter your area code, exchange, and number: ";
    cin >> userNumber.areaCode >> userNumber.exchange >> userNumber.number;
    cout << "My number is (" << myNumber.areaCode << ") "
         << myNumber.exchange << "-" << myNumber.number << endl;
    cout << "Your number is (" << userNumber.areaCode << ") "
         << userNumber.exchange << "-" << userNumber.number << endl;
    return 0;
}*/

```

```

C:\Users\itx ninja\Documents
Enter your area code, exchange, and number: 0004
50002
32126569
My number is (212) 767-8900
Your number is (4) 50002-32126569

-----
Process exited after 9.416 seconds with return value 0
Press any key to continue . . . |

```

TASK5:

```

/*task5
#include <iostream>
using namespace std;
struct point {
    int x;
    int y;
};
int main() {
    point p1, p2, p3;
    cout << "Enter coordinates for p1: ";
    cin >> p1.x >> p1.y;
    cout << "Enter coordinates for p2: ";
    cin >> p2.x >> p2.y;
    p3.x = p1.x + p2.x;
    p3.y = p1.y + p2.y;
    cout << "Coordinates of p1+p2 are: "
    << p3.x << ", " << p3.y << endl;
    return 0;
}

```

```

C:\Users\itx ninja\Documents
Enter coordinates for p1: 5.0
Enter coordinates for p2: 6.0
Coordinates of p1+p2 are: 6, 0
-----
Process exited after 25.9 seconds with return value 0
Press any key to continue

```

TASK6:

```

/*task6
#include <iostream>
using namespace std;
struct Distance {
    int feet;
    int inches;
};
struct Volume {
    Distance length;
    Distance width;
    Distance height;
};
int main() {
    Volume room = {
        {12, 6}, // Length = 12 feet 6 inches
        {10, 3}, // width = 10 feet 3 inches
        {8, 9}   // height = 8 feet 9 inches
    };
    float l = room.length.feet + room.length.inches / 12.0;
    float w = room.width.feet + room.width.inches / 12.0;
    float h = room.height.feet + room.height.inches / 12.0;
    float volume = l * w * h;
    cout << "Room dimensions:" << endl;
    cout << "Length: " << room.length.feet << " feet " << room.length.inches << " inches" << endl;
    cout << "Width : " << room.width.feet << " feet " << room.width.inches << " inches" << endl;
    cout << "Height: " << room.height.feet << " feet " << room.height.inches << " inches" << endl;
    cout << "\nVolume of room = " << volume << " cubic feet" << endl;
    return 0;
}*/

```

```

C:\Users\itx ninja\Documents
Room dimensions:
Length: 12 feet 6 inches
Width : 10 feet 3 inches
Height: 8 feet 9 inches

Volume of room = 1121.09 cubic feet

-----
Process exited after 1.424 seconds with return value 0
Press any key to continue . . .

```

TASK7:

```

int empNo;
float compensation;

: main() {
    employee e1, e2, e3;
    cout << "Enter details for Employee 1\n";
    cout << "Employee Number: ";
    cin >> e1.empNo;
    cout << "Compensation: ";
    cin >> e1.compensation;
    cout << "\nEnter details for Employee 2\n";
    cout << "Employee Number: ";
    cin >> e2.empNo;
    cout << "Compensation: ";
    cin >> e2.compensation;
    cout << "\nEnter details for Employee 3\n";
    cout << "Employee Number: ";
    cin >> e3.empNo;
    cout << "Compensation: ";
    cin >> e3.compensation;
    cout << "\n--- Employee Details ---\n";
    cout << "\nEmployee 1\n";
    cout << "Number: " << e1.empNo << endl;
    cout << "Compensation: $" << e1.compensation << endl;
    cout << "\nEmployee 2\n";
    cout << "Number: " << e2.empNo << endl;
    cout << "Compensation: $" << e2.compensation << endl;
    cout << "\nEmployee 3\n";
    cout << "Number: " << e3.empNo << endl;
    cout << "Compensation: $" << e3.compensation << endl;
    return 0;
}

```



```
C:\Users\itx ninja\Documents X + v
Enter details for Employee 1
Employee Number: zain
Compensation:
Enter details for Employee 2
Employee Number: Compensation:
Enter details for Employee 3
Employee Number: Compensation:
--- Employee Details ---

Employee 1
Number: 0
Compensation: $0

Employee 2
Number: 1
Compensation: $0

Employee 3
Number: -1
Compensation: $nan

-----
Process exited after 8.508 seconds with return value 0
Press any key to continue . . . |
```

TASK8:

```
C:\Users\itx ninja\Documents X + v
Enter date (MM/DD/YYYY):
11/14/2006
Date is: 11/14/2006

-----
Process exited after 12.21 seconds with return value 0
Press any key to continue . . . |
```

TASK9:

```

/*task9
#include <iostream>
using namespace std;
struct time {
    int hours;
    int minutes;
    int seconds;
};
int main() {
    time t1;
    cout << "Enter time (HH:MM:SS): ";
    char ch1, ch2;
    cin >> t1.hours >> ch1 >> t1.minutes >> ch2 >> t1.seconds;
    long totalsecs;
    totalsecs = t1.hours * 3600 + t1.minutes * 60 + t1.seconds;
    cout << "Total seconds = " << totalsecs << endl;
    return 0;
}*/

```

```

C:\Users\itx ninja\Documents >
Enter time (HH:MM:SS): 20:15:0.5
Total seconds = 72900

-----
Process exited after 19.58 seconds with return value 0
Press any key to continue . . . |

```

TASK10:

```

C:\Users\itx ninja\Documents >
Enter first time (HH:MM:SS): 24:16:5
Enter second time (HH:MM:SS): 12:15:6
Sum of time = 36:31:11

-----
Process exited after 24.55 seconds with return value 0
Press any key to continue . . . |

```

HOME TASK

```

//////////////////////////////////////
/*TASK1
#include <iostream>
using namespace std;
struct hotel {
    string name;
    int dishes;
    float bill;
};
int main() {
    hotel h[3];
    cout << "Enter details of last 3 hotels visited:\n";
    for(int i = 0; i < 3; i++) {
        cout << "\nHotel " << i + 1 << endl;
        cout << "Hotel name: ";
        cin >> h[i].name;
        cout << "Number of dishes: ";
        cin >> h[i].dishes;
        cout << "Total bill: ";
        cin >> h[i].bill;
    }
    cout << "\n--- Records with 15% Tax Added ---\n";
    float grandTotal = 0;
    for(int i = 0; i < 3; i++) {
        float taxBill = h[i].bill + (h[i].bill * 0.15);
        grandTotal += taxBill;
        cout << "\nHotel Name: " << h[i].name << endl;
        cout << "Dishes: " << h[i].dishes << endl;
        cout << "Bill after 15% tax: " << taxBill << endl;
    }
    cout << "\nTotal Bill of all hotels (with tax): " << grandTotal << endl;
}

```

TASK1: return 0;

C:\Users\itx ninja\Documents X

+

▼

Enter details of last 3 hotels visited:

Hotel 1

Hotel name: NISHAT

Number of dishes: STEAK

Total bill:

Hotel 2

Hotel name: Number of dishes: Total bill:

Hotel 3

Hotel name: Number of dishes: Total bill:

--- Records with 15% Tax Added ---

Hotel Name: NISHAT

Dishes: 0

Bill after 15% tax: 0

Hotel Name:

Dishes: 1

Bill after 15% tax: 0

Hotel Name:

Dishes: 4254245

Bill after 15% tax: 0

Total Bill of all hotels (with tax): 0

Process exited after 16.18 seconds with return value 0

Press any key to continue . . . |

```

/*TASK2
#include <iostream>
using namespace std;
struct Person {
    float income;
    float taxRate;
    float tax;
};
int main() {
    Person p[5];
    cout << "Enter income and tax rate for 5 persons:\n";
    for(int i = 0; i < 5; i++) {
        cout << "\nPerson " << i + 1 << endl;
        cout << "Income: ";
        cin >> p[i].income;
        cout << "Tax rate (%): ";
        cin >> p[i].taxRate;
        p[i].tax = (p[i].income * p[i].taxRate) / 100;
    }
    cout << "\n--- Tax Payable for Each Person ---\n";
    for(int i = 0; i < 5; i++) {
        cout << "Person " << i + 1 << " Tax: " << p[i].tax << endl;
    }
    return 0;
}*/

```

TASK2:

C:\Users\itx ninja\Documents

+

▼

Person 1

Income: 45000

Tax rate (%): 5

Person 2

Income: 60000

Tax rate (%): 8

Person 3

Income: 100000

Tax rate (%): 10

Person 4

Income: 150000

Tax rate (%): 15

Person 5

Income: 200000

Tax rate (%): 20

--- Tax Payable for Each Person ---

Person 1 Tax: 2250

Person 2 Tax: 4800

Person 3 Tax: 10000

Person 4 Tax: 22500

Person 5 Tax: 40000

Process exited after 41.07 seconds with return value 0

Press any key to continue . . .